

WASP-50b

R-1487

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_211212 · created 2026-08-01T13:30:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459560.6888 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1751 +/- 0.0063
Transit depth (Rp/Rs)^2	3.06 +/- 0.22 [%]
Orbital Inclination (inc)	84.98 +/- 0.55
Airmass coefficient 1 (a1)	1.037 +/- 0.011
Airmass coefficient 2 (a2)	-0.025 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#2 - [373, 123]
Optimal Aperture	3.88
Optimal Annulus	15.95
Transit Duration (day)	0.0808 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1751 $\pm$ 0.0063 vs 0.1404 (deviation 5.47 σ)
Duration fit vs published	0.0808 d vs 0.0754 d (deviation 1.37 σ)
Mid-time O-C	1.4 min
Depth SNR	27.6
Residual scatter	1.08 %

Light curve

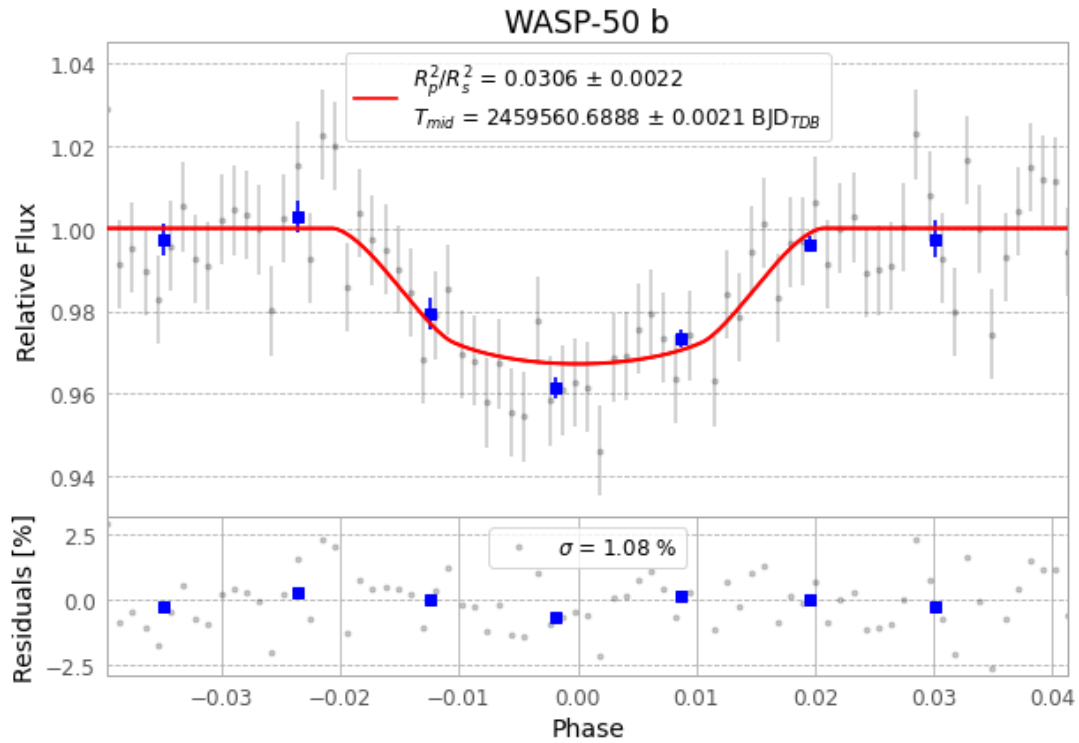


Figure 1 — FinalLightCurve_WASP-50 b_2021-12-11.png (SHA-256 c08cd7f5f98b06aa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 146], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 25921f1b6a4fe03e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-50211212023312.FITS → WASP-50211212062112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-211212.log (c63740f3a3d8...), FinalLightCurve_WASP-50 b_2021-12-11.png (c08cd7f5f98b...), FinalLightCurve_WASP-50 b_2021-12-11.csv (fd3e62f56543...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 13:30Z.